

YG1006 Technical Data Sheet

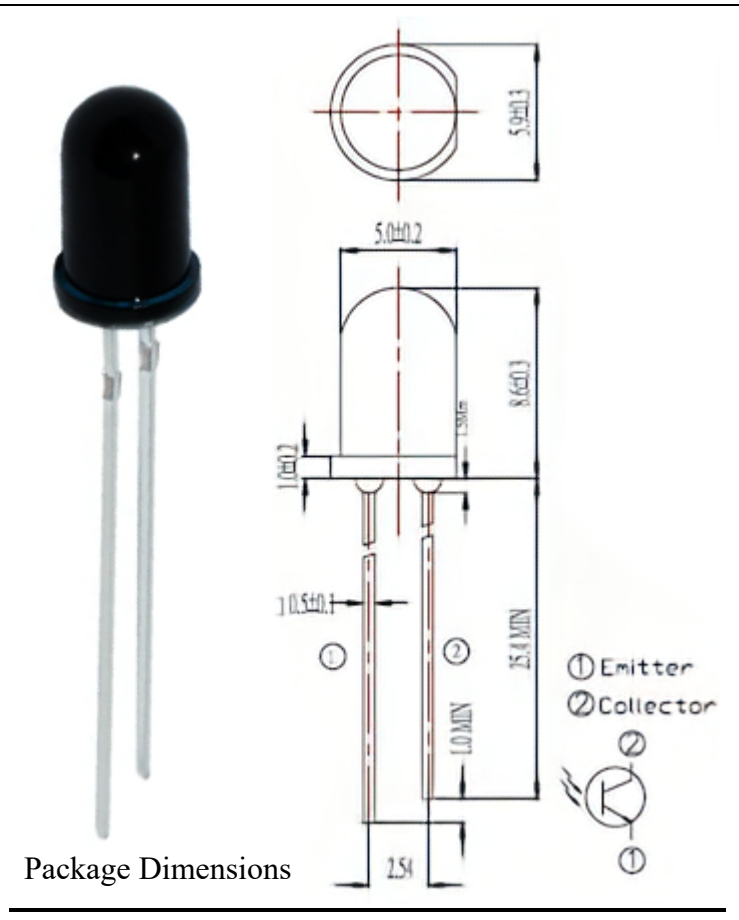
Features

- Fast response time
- High photo sensitivity
- Pb free
- This product itself will remain within RoHS compliant version.

Descriptions

- YG1006 is a high speed and high sensitive NPN silicon phototransistor a standard 5mm package.

Due to its black epoxy the device is sensitive to infrared radiation.



Note : 1. All dimensions are in millimeters

2. Tolerances unless dimensions $\pm 0.25\text{mm}$

Absoulute Maximum Ratings (Ta=25°C)

Parameter	Symbol	rating	units
Collector-Emitter Voltage	V_{CE0}	30	V
Emitter-Collector-Voltage	V_{ECO}	5	V
Collector Current	I_c	20	mA
Operating Temperature	T_{opr}	-25~+85°C	°C
Storage Temperature	T_{stg}	-40+85°C	°C
Lead Soldering Temperature	T_{sol}	260	°C
Power Dissipation at (or below) 25°C FreeAir Temperature	P_c	75	mW

Electro-Opticai Characteristics (Ta=25°C)

Parameter	Symbol	Condition	Min.	Typ.	Max.	Units
Collector-Emitter Breakdown Voltage	BV_{CEO}	$I_C=100\mu A$ $E_e=0mW/C\ m^2$	30	---	---	V
Emitter-Collector Saturation Voltage	BV_{ECO}	$I_E=100\mu A$ $E_e=0mW/C\ m^2$	5	---	---	V
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$	$I_C=2mA$ $E_e=1mW/C\ m^2$	---	---	0.4	V
Rise Time	t_f	$V_{CE}=5V$ $I_C=1mA$ $R_L=1000\Omega$	---	15	---	μS
Fall Time	t_f		---	15	---	
Collector Dark Current	I_{CEO}	$E_e=0mW/C\ m^2$ $V_{CE}=20V$	---	----	100	nA
On State Collector Current	$I_C(on)$	$E_e=1mW/C\ m^2$ $V_{CE}=5V$	1.77	----	7.07	mA
Wavelength of Peak Sensitivity	λ_P	---	---	940	---	nm
Rang of Spectral Bandwidth	$\lambda_{0.5}$	---	---	760-1100	---	nm

Rankings

Parameter	Symbol	Min	Max	Unit	Test Condition
	$I_C(on)$			mA	$V_{CE}=5V$ $E_e=1mW/C\ m^2$
J		1.77	3.61		
K		2.67	5.07		
L		4.18	7.07		

Typical Electro-Optical Characteristics Curves

Fig.1 Collector Power Dissipation vs.
Ambient Temperature

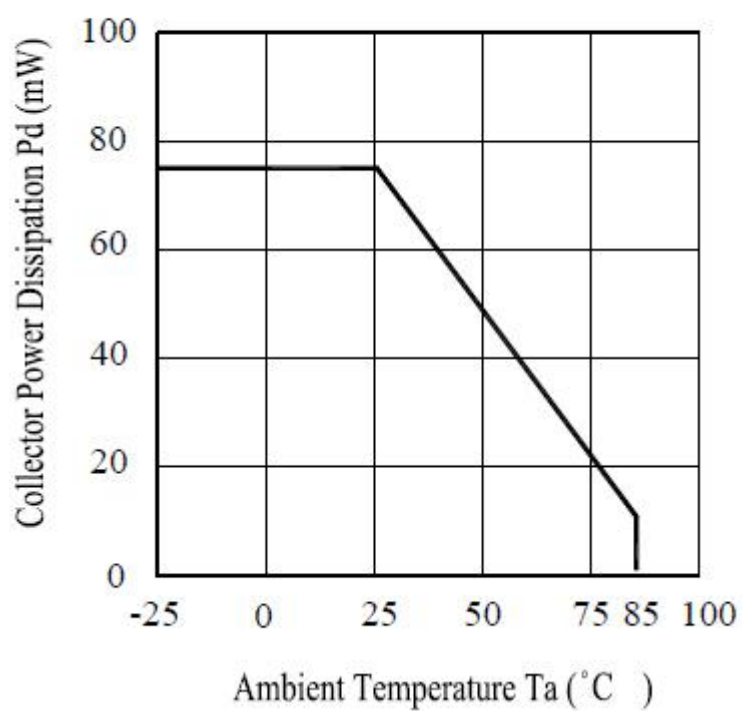


Fig.2 Spectral Sensitivity

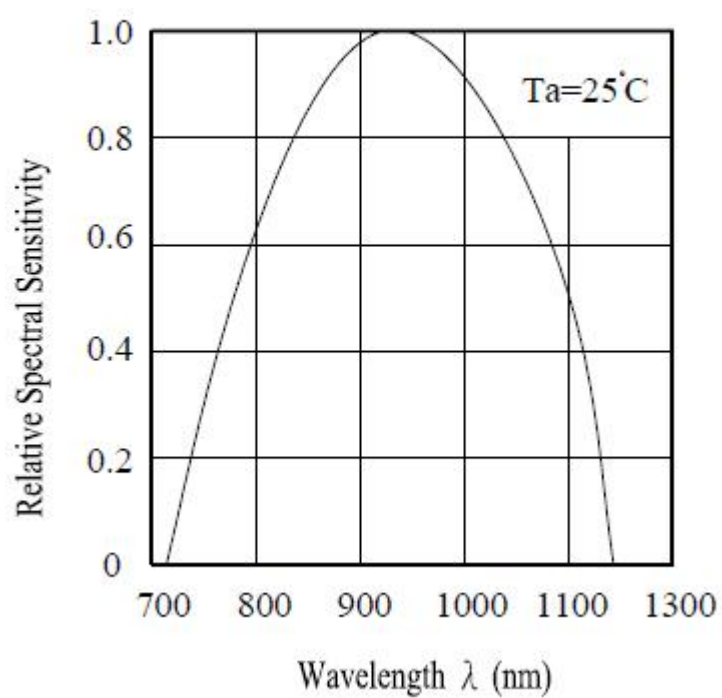


Fig.3 Relative Collector Current vs.
Ambient Temperature

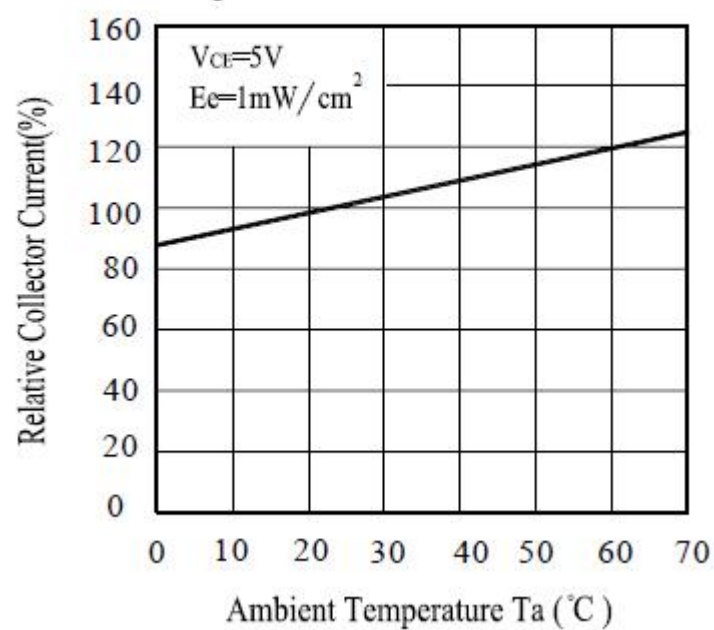
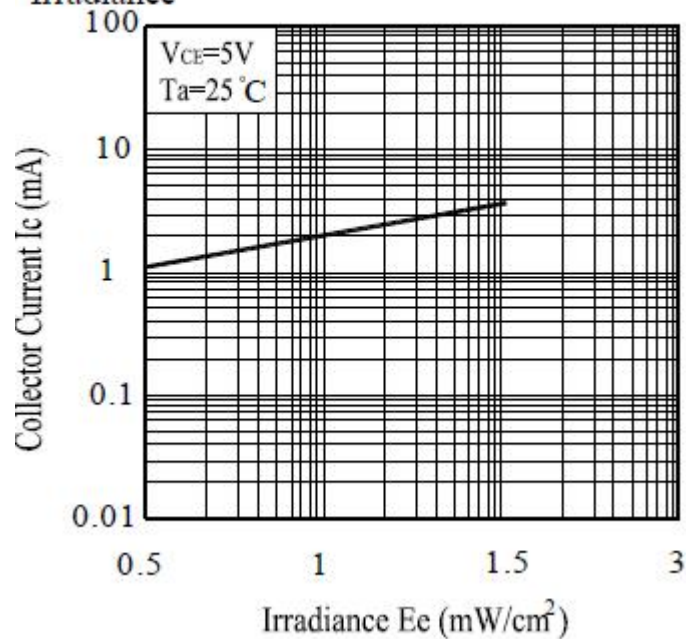


Fig.4 Collector Current vs.
Irradiance



Typical Electro-Optical Characteristecs

Fig.5 Collector Dark Current vs.
Ambient Temperature

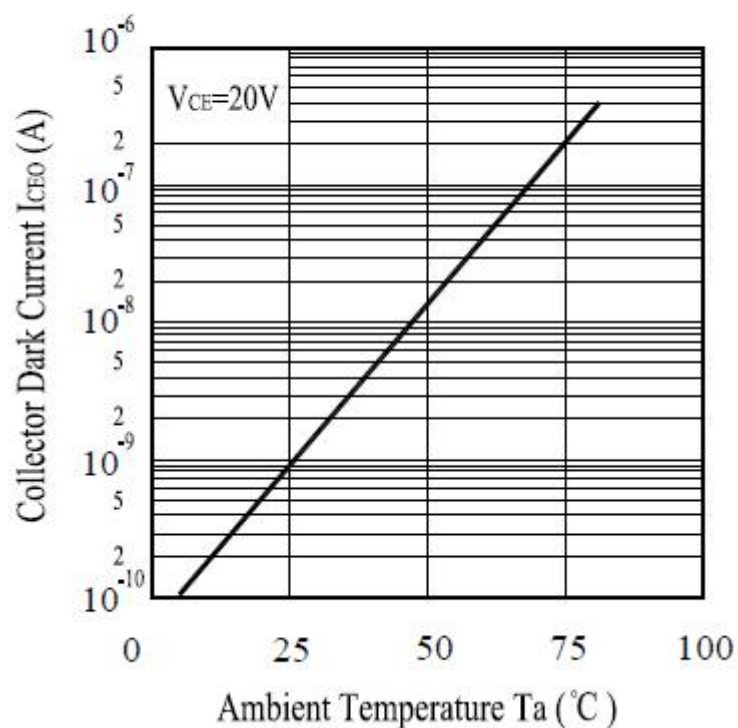
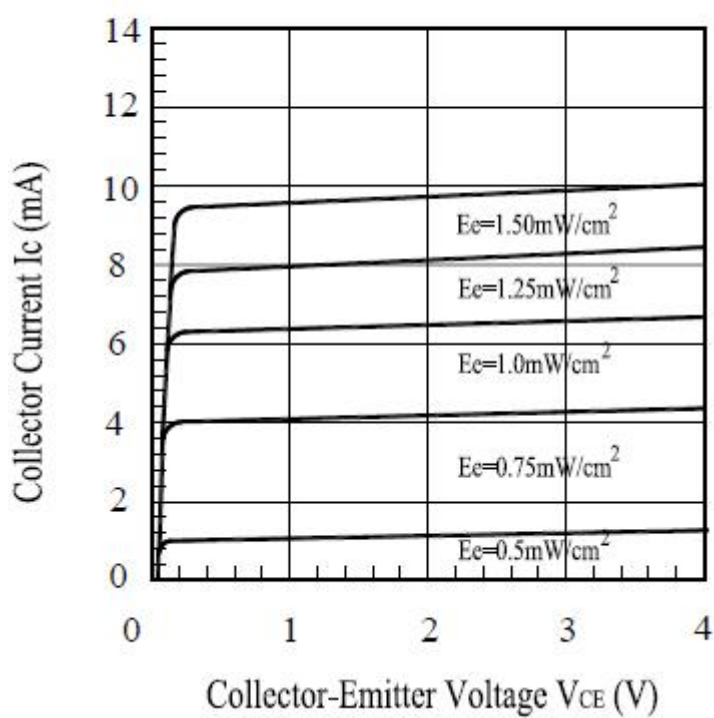


Fig.6 Collector Current vs.
Collector-Emitter Voltage



Reliability Test Item And Condition

The reliability of products shall be satisfied with items listed below.

Confidence level : 90%

LTPD : 10%

NO.	Item	Test Conditions	Test Hours/ Cycles	Sample Sizes	Failure Judgement Criteria	Ac/Re
1	Solder Heat	TEMP. : $260^{\circ}\text{C} \pm 5^{\circ}\text{C}$	10secs	22pcs	$I_{C(ON)} \leq L \times 0.8$ L : the initial test value	0/1
2	Temperature Cycle	H : $+100^{\circ}\text{C}$ 15mins $\updownarrow$ 5mins L : -40°C 15mins	300Cycles	22pcs		0/1
3	Thermal Shock	H : $+100^{\circ}\text{C}$ 5mins $\updownarrow$ 10secs L : -10°C 5mins	300Cycles	22pcs		0/1
4	High Temperature Storage	TEMP. : $+100^{\circ}\text{C}$	1000hrs	22pcs		0/1
5	Low Temperature Storage	TEMP. : -40°C	1000hrs	22pcs		0/1
6	DC Operating Life	$V_{CE}=5\text{V}$	1000hrs	22pcs		0/1
7	High Temperature/ High Humidity	85°C / 85% R.H	1000hrs	22pcs		0/1